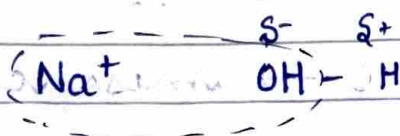


Unit → 3

→ Ion-dipole interaction

- It occurs between a fully charged ion and a neutral polar molecule that possesses the permanent dipole moment.



$$V \propto \frac{1}{r^2}$$

- Higher ionic charge gives stronger interaction
- Polarity of solvent ↑ interaction ↑

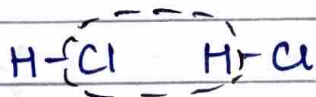
→ Vander - waal Forces

These forces are due to attractions between neutral molecules and are very weak than ionic / covalent bond.

- Dipole - Dipole interaction

→ Between 2 polar molecules

$$E \propto \frac{1}{r^6}$$



- Dipole - induced dipole

A polar molecule approaches a non polar molecule, distorting its electron cloud & induce a temporary dipole.

→ Potential energy surfaces

It's mathematical/graphical energy representation of a molecule as a function of its geometry. It's based on Born-Oppenheimer approximation which assumes that because nuclei are much heavier than electrons, they move more slowly. This allows us to calculate electronic energy for fixed nuclear positions. As we vary these positions, energy changes creating a surface.

For a system of N atoms, the potential energy states has $3N-6$ degrees of freedom eg: H_2

• For diatomic molecule like H_2

only variable is the bond length r and PES is a simple curve

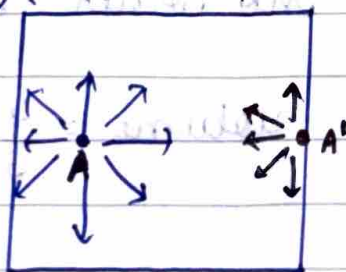
$$V(r) = D_e (1 - e^{-a(r-r_e)})^2$$

→ Vander Waal's equation for state of Real gases

A gas which follows $PV = nRT$ at all temp & pressure is called as ideal gas. Vander Waal modified this ideal gas equation by making pressure and volume corrections.

1. Correction for pressure

Let's consider a molecule A of gas placed inside a container as shown. This molecule A is attracted by other molecules in all direction with same force and net force acting on it is 0.



But when it strikes the wall of container (A') it is pulled backwards by other molecules and velocity & momentum

with which it strikes the wall would be less than that of which it will strike in the absence of attractive forces. This reduction in momentum results in absence of pressure. The observed pressure is less than actual pressure.

The correction of pressure depends upon no. of molecules striking a unit area per unit time of walls of container and resultant inward pull of colliding molecules which are dependent of density of gas P . so correction of pressure (p)

$$p \propto p^2$$

$$p \propto \frac{1}{V}$$

$$\Rightarrow p \propto \frac{1}{V^2}$$

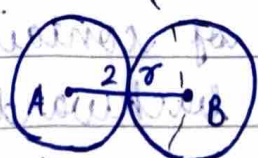
$$\Rightarrow p = \frac{a}{V^2}$$

Hence corrected pressure is

$$p + p = P + \frac{a}{V^2}$$

2. Volume correction

Due to finite size of gas molecule, the actual space available for the movement of molecule is less than volume of vessel. The molecules have sphere of influence around them of radius $2r$ within which no other molecule can penetrate.



$$\text{Volume} = \frac{4}{3} \pi r^3 = \frac{\pi}{3} (2r)^3$$

sphere of influence

The center of any two molecules can approach each other only by a minimum distance $2r$. The volume of sphere of influence of each molecule is

$$S = \frac{4}{3} \pi (2r)^3 = 8 \pi r^3$$

Let's fill the whole space by molecules of volume V with n molecules one by one

volume for 1st molecule = $V - S$

volume for 2nd molecule = $V - 2S$

volume for n th molecule = $V - (n-1)S$

Hence, average space for each molecule is

$$\frac{V + V - S + V - 2S + \dots + V - (n-1)S}{n}$$

$$= \frac{nV - S [1 + 2 + 3 + \dots + (n-1)]}{n}$$

$$= V - S \frac{(n-1)(n)}{2n}$$

$$= V - \frac{nS}{2} + \frac{S}{2}$$

As no. of molecules is very large, we neglect $S/2$, so average space for each molecule is $V - \frac{nS}{2}$

$$= V - \frac{n}{2} (8\pi r^3) = V - 4n\pi r^3$$

$$= V - b$$

Hence, using ideal gas equation

$$\left(P + \frac{a}{V^2}\right) (V - b) = nRT$$

→ Critical Phenomena

All gases can be liquified by decreasing temperature and increasing pressure because $T \downarrow$ $KE \downarrow$ and as $P \uparrow$ gas molecules come closer to each other and Force of attraction increases and gas liquifies.

- **Critical Temperature**
Temp at which any gas can be liquified but above this gas can't be liquified even by applying very high pressure gas can be liquified below this temp by applying pressure.

- **Critical Pressure**
Required pressure for liquifying gas at a critical temp.

$$\left(P_c + \frac{a}{V_c^2} \right) (V_c - b) = RT_c$$

differentiate w.r.t. V_c

$$P_c = \frac{RT_c}{V_c - b} - \frac{a}{V_c^2} \quad \text{--- (1)}$$

$$\frac{dP_c}{dV_c} = \frac{-RT_c}{(V_c - b)^2} + \frac{2a}{V_c^3} \quad \text{--- (2)}$$

At critical temp;

$$\frac{dP_c}{dV_c} = 0$$

$$\Rightarrow \frac{2a}{V_c^3} = \frac{RT_c}{(V_c - b)^2} \quad \text{--- (3)}$$

Diff ② wrt V_c once again

$$\frac{d^2 P_c}{d V_c^2} = \frac{2RT_c}{(V_c - b)^3} - \frac{6a}{V_c^4}$$

$$\text{at } T_c \quad \frac{d^2 P_c}{d V_c^2} = 0$$

$$\Rightarrow \frac{2RT_c}{(V_c - b)^3} = \frac{6a}{V_c^4}$$

$$\Rightarrow \frac{3a}{V_c^4} = \frac{RT_c}{(V_c - b)^3} \quad \text{--- (4)}$$

$$\text{(3)} \div \text{(4)} : \frac{2RT_c}{(V_c - b)^3} \div \frac{RT_c}{(V_c - b)^3} = \frac{6a}{V_c^4} \div \frac{3a}{V_c^4}$$

$$\frac{2a}{V_c^3} = \frac{RT_c}{(V_c - b)^2}$$

$$\frac{2V_c}{V_c^3} = \frac{V_c - b}{V_c^2} \Rightarrow V_c - b = 2b$$

$$2V_c = 3V_c - 3b$$

$$2V_c = 3V_c - 3b \Rightarrow V_c = 3b$$

Put in ③ $V_b + U_b = V$

$$\frac{2a}{27b^3} = \frac{RT_c}{(3b - b)^2} = \frac{RT_c}{4b^2}$$

$$2a = 27b^3 \cdot \frac{RT_c}{4b^2} = 27b \cdot \frac{RT_c}{4} = 10b$$

$$\Rightarrow T_c = \frac{8a}{27Rb}$$

$$\underline{\underline{\frac{8a}{27Rb}}}$$

Put in ①

$$P_c = \frac{8a}{27b(2b)} - \frac{a}{9b^2}$$

$$P_c = \frac{4a}{27b^2} - \frac{3a}{27b^2}$$

$$\Rightarrow \underline{P_c = \frac{a}{27b^2}}$$

→ Nernst equation

As we know

$$G_1 = H_1 - TS_1$$

Let at temp T , condition is isothermal

Hence $G_1 = H_1 - TS_1$ (initially) and

$$G_2 = H_2 - TS_2 \text{ (finally)}$$

Now subtract these equations

$$G_2 - G_1 = H_2 - H_1 - T(S_2 - S_1)$$

$$\Rightarrow \Delta G = \Delta H - T\Delta S$$

Also, $H = U + PV$

$$\Rightarrow G = U + PV - TS \Rightarrow \text{diff B.S}$$

$$\Rightarrow dG = dU + PdV + VdP - TdS - SdT$$

Also, $dS = \frac{dq}{T} \Rightarrow TdS = q$

also $q = dU + PdV$

$$TdS = dU + PdV$$

$$\Rightarrow dG = \cancel{dU} + PdV + VdP - \cancel{dU} - \cancel{PdV} - SdT$$

$$\Rightarrow \underline{dG = VdP - SdT}$$

Now, at constant temp T , $dT = 0$

$$\Rightarrow dG = v dP$$

For 1 mole ideal gas

$$dG = \frac{RT}{P} dP$$

$$\Rightarrow \int_{G^\circ}^G dG = RT \int_{P_0}^P \frac{1}{P} dP$$

$$\Rightarrow G - G^\circ = RT \ln \frac{P}{P_0} = RT \ln P$$

For n moles

$$nG = nG^\circ + nRT \ln P$$



$$\Delta G = (\sum G)_{\text{product}} - (\sum G)_{\text{reactants}}$$

$$\Rightarrow Q = \frac{(P_C)^c (P_D)^d}{(P_A)^a (P_B)^b} = Q$$

$$\Rightarrow nG = nG^\circ + nRT \ln Q$$

$$\Rightarrow \Delta G = \Delta G^\circ + RT \ln Q$$

Also,

$$\Delta G = -nF E_{\text{cell}} \quad \& \quad \Delta G^\circ = -nF E^\circ_{\text{cell}}$$

$$\Rightarrow -nF E_{\text{cell}} = -nF E^\circ_{\text{cell}} + RT \ln Q$$

$$\Rightarrow E_{\text{cell}} = E^\circ_{\text{cell}} - \frac{RT}{nF} \ln Q$$

$$\Rightarrow E_{\text{cell}} = E^\circ_{\text{cell}} - \frac{2.303 RT}{nF} \log_{10} Q$$

↪ Nernst equation

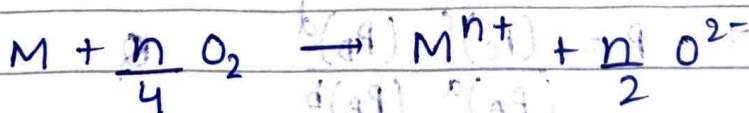
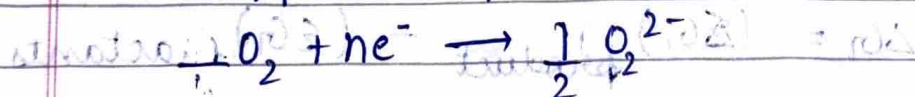
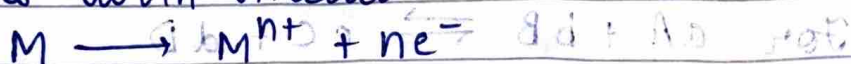
- corrosion gradual disintegration of metal by chemical action of environment leading to weakening of metal upto several layers.

Tendency of metal to convert from pure high energy form to stable low energy complex leads to corrosion.

• Dry corrosion

- direct action of O_2 on metal in absence of moisture

- O_2 from air is attracted & absorbed to metal surface due to Vander waal forces & reacts with metal



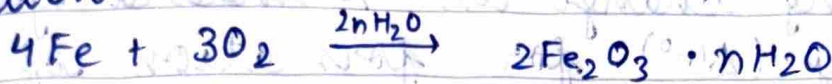
- Gases like H_2S , SO_2 , CO_2 , Cl_2 also attack on metal surfaces & lead to corrosion.

Extent of corrosion depends upon the chemical reactivity between metal & gas

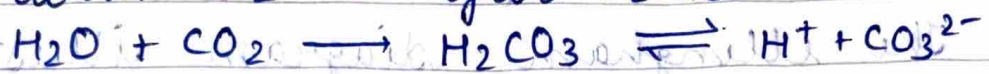
- H_2 can get into interstitial spaces and combine to form H_2 gas leading to cracking & blistering of metals

- Chemical attack of liquid metal over metal/alloy, corrodes it & makes it weaker.

- Wet corrosion (Rusting of iron) occurs due to flow of e^- from anode to cathode regions of metals via conducting solution



- Metal surface has water present on it which reacts with CO_2 to give H_2CO_3



- Oxidation takes place at cracks on metal surfaces & creates anodic areas leading to oxidation

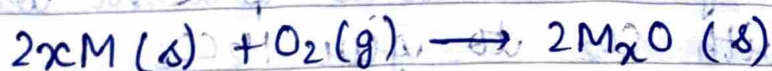
- Now e^- move to adjacent wet area which behaves like cathode & these reactions combine to form rust.

- Applications of Nernst equation

- Determination of cell potential
- Finding EMF of cell at unstandard condition
- Finding voltage due to concentration cells
- Finding equilibrium constant
- Determination of unknown ionic concentration
- Finding solubility product
- In physiology & microbiology
- In potentiometric titrations

→ Ellingham Diagram

It is a graph that shows the temperature dependence on stability of compounds by plotting change in standard Gibbs free energy (ΔG°) for formation of metal oxides against temperature (T).



The Ellingham diagram illustrates how the tendency of this metal oxide to get reduced changes as temperature increases.

A sharp increase in slope of this graph indicated phase change i.e. melting/boiling.

→ APPLICATIONS OF ELLINGHAM DIAGRAMS IN METALLURGY

1. Selection of Reducing agent
A metal can reduce the oxide of any other metal that appears above it in Ellingham diagram as ΔG° for combined reaction will be negative.
2. Predicting the temperature of reduction
This diagram shows a specific temperature at which reducing agent becomes effective.
3. Predicts ease of reduction
The higher a metal is in Ellingham diagram, the less stable it is and hence easy to reduce.

4. Preventing re-oxidation

The diagram helps to determine partial pressure of oxygen in equilibrium with a metal & its oxide. This is vital in refining process to ensure purified metal does not react back.

→ Acid-Base Equilibria

(TFTB)

assemblage

... of ... for ... at ...
... is ... at ...

... of ... is ...
... is ...
... is ...

assemblage

... is ...

→ Water Chemistry

• Sources of water

Rain, river, sea, underground

→ Impurities

Physical colour, turbidity, taste, odour

Chemical chemicals, acidity, gases, minerals

Biological microorganisms

→ Hardness

(झट्ट)

Property to prevent lathering of soaps, more water is required to produce less lather

Hard water contains salts of Ca & Mg which displace Na ions to form stearates of their own which is insoluble & hence prevents lathering

• Temporary Hardness

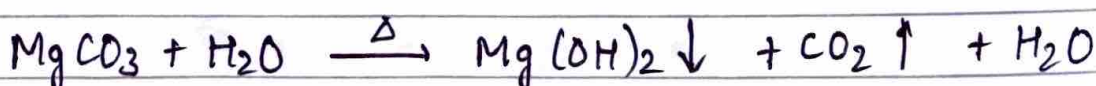
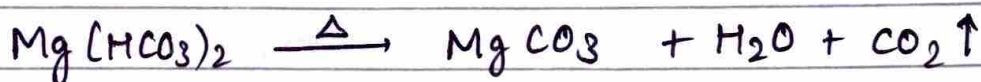
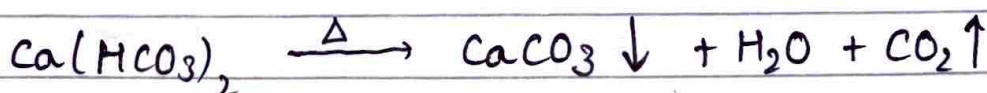
→ due to CO_3^{2-} or HCO_3^-

→ Removed by boiling $\because$ HCO_3^- on boiling gives CO_3 which is insoluble & CO_2 rises up & H_2O separates

• Permanent Hardness

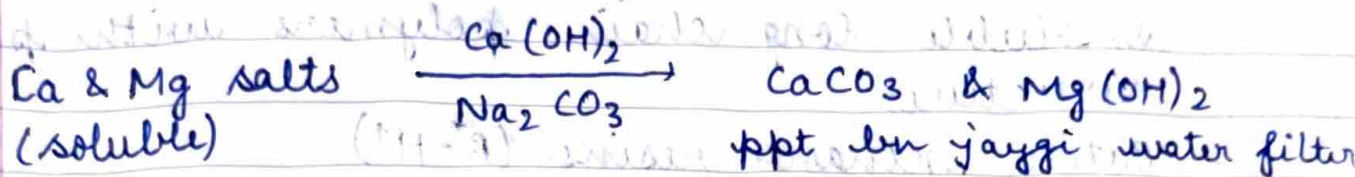
→ due to Cl^- & SO_4^{2-}

→ Can't be removed



→ Removal of Permanent hardness

• Lime soda process



• zeolite process

contains sodium zeolite. If impurity is CaCl_2 it reacts with Na_2Ze to form calcium zeolite and NaCl (Hardness)

→ Limitations

- If water is turbid, it must be coagulated else it would clog pores of zeolite bed.
- If water has large coloured ions, their zeolite would be produced and can't be regenerated like sodium zeolite.
- If water has mineral acid, they must be neutralised also zeolite bed is destroyed.

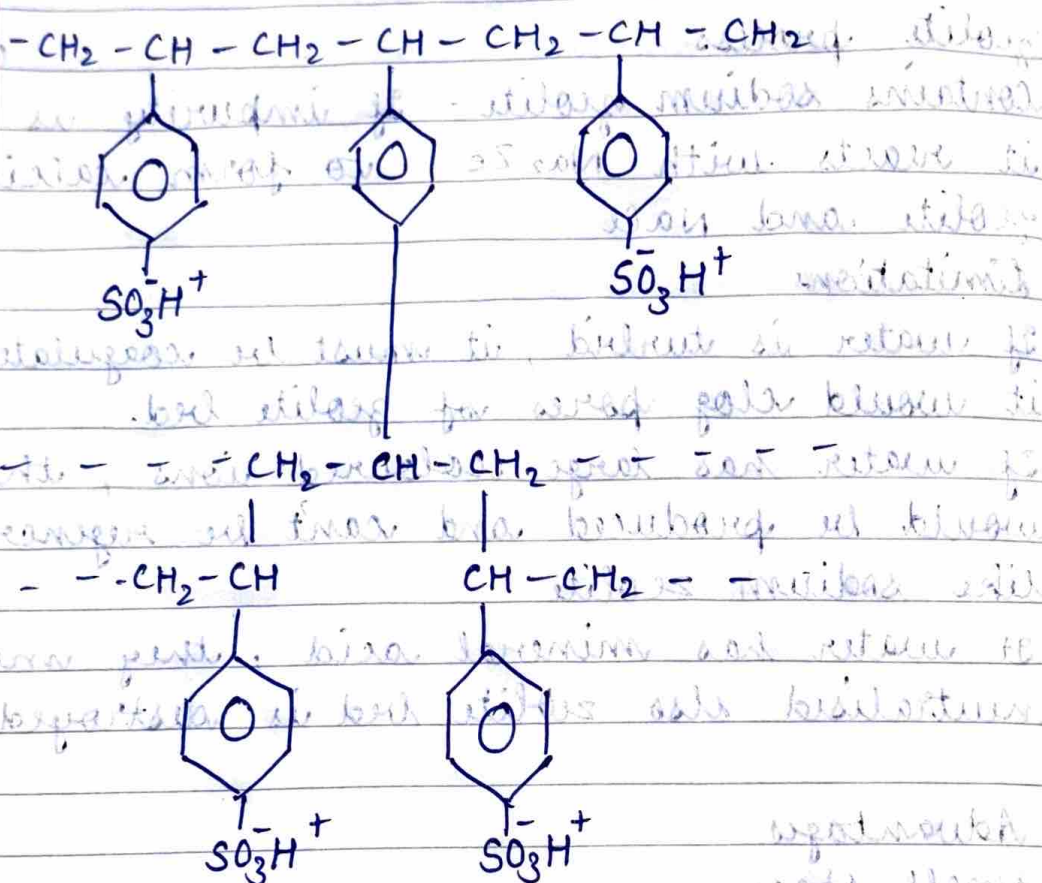
→ Advantages

- Small space
- Almost hardness is removed
- No sludge formation/precipitation
- Clean process
- Time saving
- Less maintenance

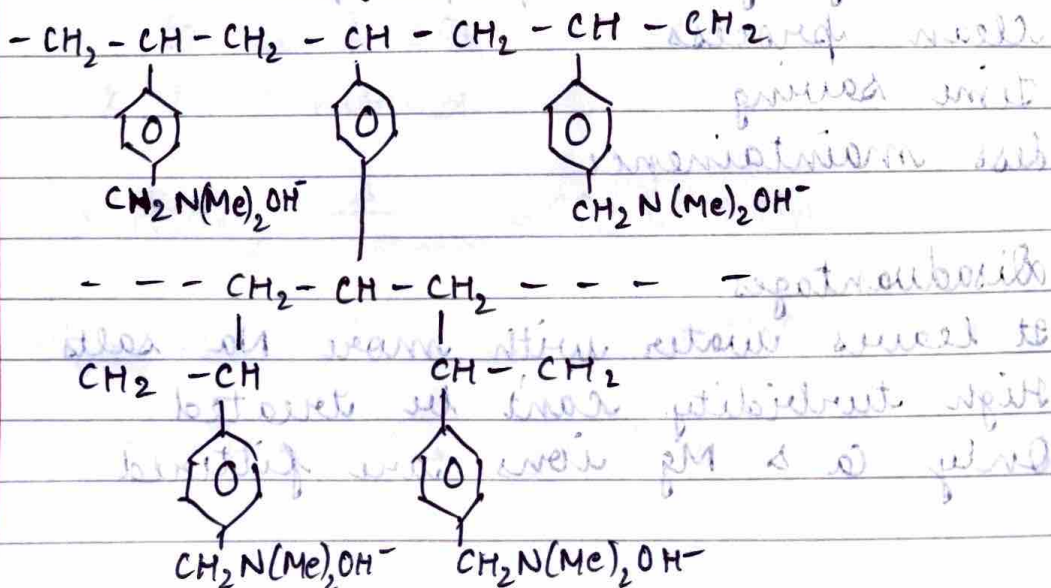
→ Disadvantages

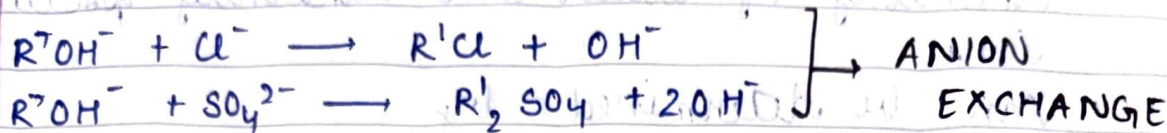
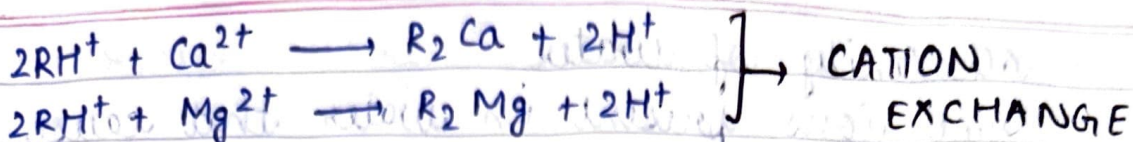
- It leaves water with more Na salts
- High turbidity can't be treated
- Only Ca & Mg ions are filtered

- Ion-exchange Process
It uses ion exchange resins which contains insoluble long chain polymers with porous structures
→ Cation exchange resins (R^+-H^+)



- Anion exchange Resins (R^+-OH^-)





For Regeneration



→ Advantage

- Soften highly acidic & alkaline water
- Produce water of low hardness (2 ppm)
- Good to treat water for boilers

→ Disadvantage

- Costly & expensive chemicals & equipments
- Turbidity ↑ water can't be used

Units of hardness

ppm, mg/l, ° French, ° Clark

$CaCO_3$ is used as standard for estimating the hardness because it provides a uniform scale for comparison & its molar mass makes calculations easier. It is virtually insoluble and precipitates out.

→ Alkalinity of water

Some ions present in water that cause the alkalinity of water are alkalinity positive ions

Can be determined by

→ Acid base titration

→ Phenolphthalein & methyl orange indicators



Add phenolphthalein, it gives pink color, titrate with acid, colour disappears, note volume (V_1)

Add methyl orange, it gives yellow colour, titrate, colour disappears & note

Hard water
(V)

volume (V_2)

$$N_1 V = N_2 V_1$$

$$N_1 V = N_3 V_2$$

$$P = \frac{N_2 \times \text{Eq. wt.} \times 1000}{(\text{CaCO}_3)}$$

$$M = \frac{N_3 \times \text{Eq. wt.} \times 1000}{(\text{CaCO}_3)}$$

Alkalinity

OH^-

CO_3^{2-}

HCO_3^-

$$P = 0$$

$$0$$

$$0$$

$$M$$

$$P = 1/2 M$$

$$0$$

$$2P/M$$

$$0$$

$$P = M$$

$$P/M$$

$$0$$

$$0$$

$$P > M/2$$

$$2P - M$$

$$2(M - P)$$

$$0$$

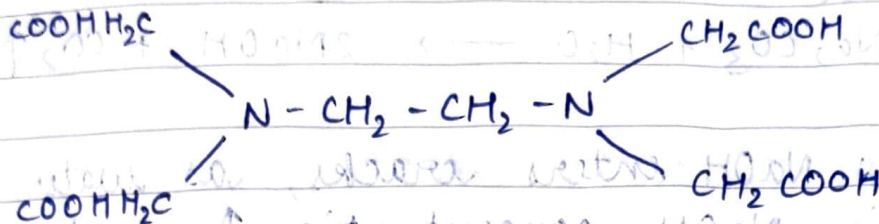
$$P < M/2$$

$$0$$

$$2P$$

$$M - 2P$$

→ Determination of Permanent hardness by EDTA



We do complexometric titrations using EDTA and EBT.

- Buffer solution of $\text{NH}_4\text{OH} + \text{NH}_4\text{Cl}$ is used to insure completion of reaction by maintaining pH at 10.

- EBT added to hardwater, forms weak wine red complex with metal ions.
- When EDTA is added, it displaces the EBT and forms metal-EDTA complex. Once all metal are bound to EDTA, EBT sets free & returns to steel blue colour.

$$\text{Hardness} = \frac{V_1 \times M \times 100 \times 1000}{V_2} \text{ ppm}$$

$V_1 \rightarrow$ Volume of EDTA used in

$M \rightarrow$ molarity of EDTA

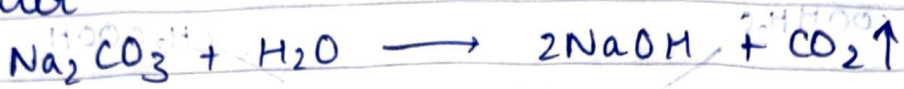
$V_2 \rightarrow$ Volume of water sample

$100 \rightarrow$ M.M. of CaCO_3

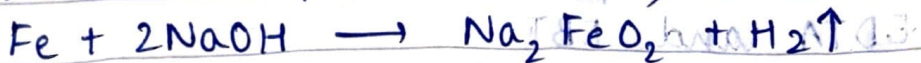
→ Caustic embrittlement

It's specific form of stress corrosion cracking that occurs in boilers where metal becomes brittle and cracks due to NaOH . It was major cause of boiler explosions in early times.

ATQ In many boilers Na_2CO_3 is added to soften water



This NaOH enters cracks, as water evaporates this NaOH concentration $\uparrow$ and reacts with iron (from boilers)



makes metal brittle

→ Requirements

- High stress due to internal pressure
- $[\text{NaOH}] \uparrow \uparrow$
- Tiny gaps must exist

→ Sludge & Scale

These are deposits formed inside boilers when hard water is heated

Sludge is soft, slimy, loose ppt that floats in water or settle down where flow is low. Formed due to substances more soluble in hot water such as MgCl_2 , MgSO_4 . It can be easily removed. It causes foaming & damage turbine blades & interfere heat flow.

Scale is hard crystalline coating that sticks to inner coating of boilers. Formed due to substances more soluble in cold water CaCO_3 , CaSO_4 . It leads to fuel wastage, boiler explosions & corrosion.